# 30 Python Libraries to (Hugely) Boost Your Data Science Productivity

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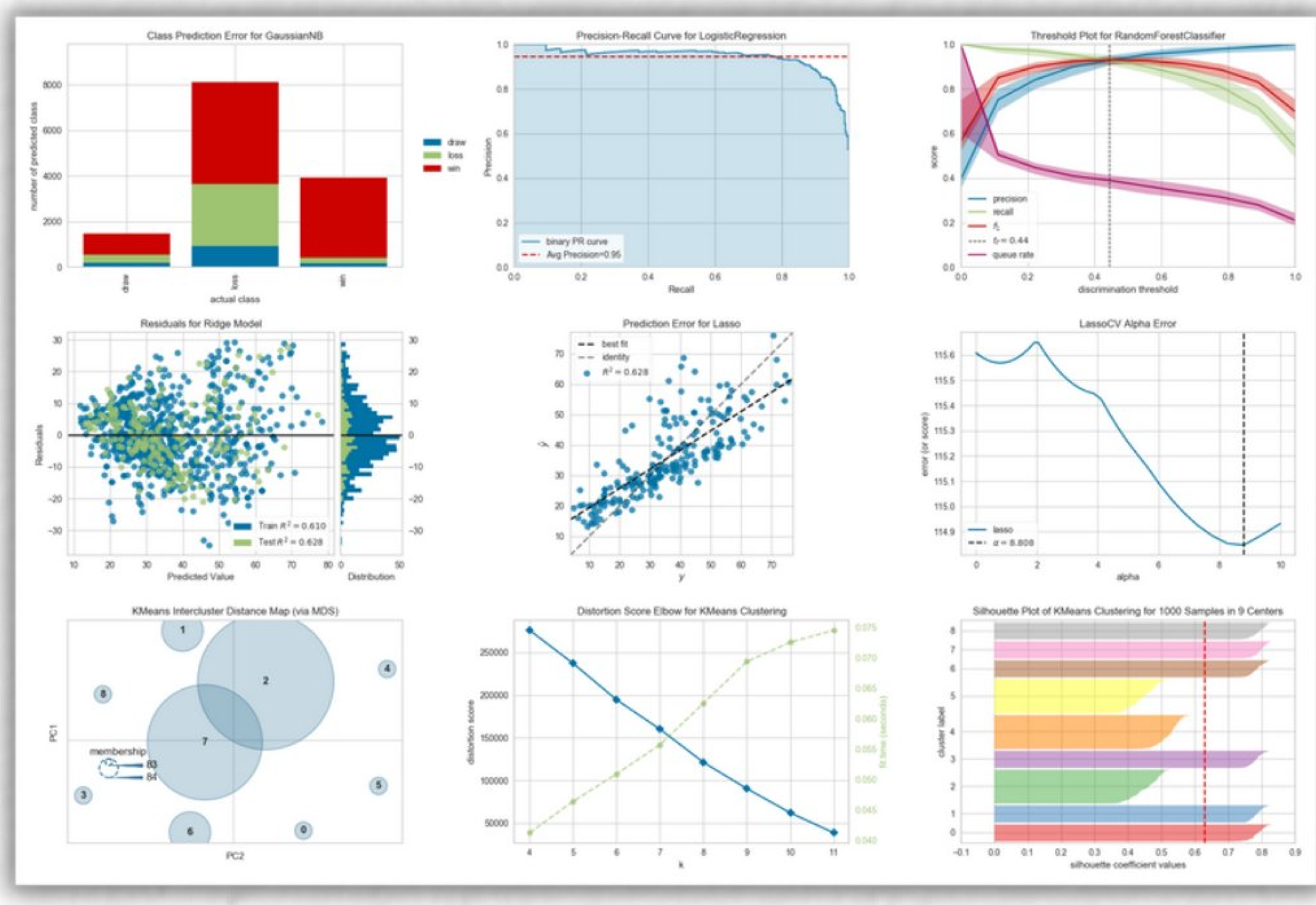
**Data Science is much more than Pandas, NumPy and Sklearn.**

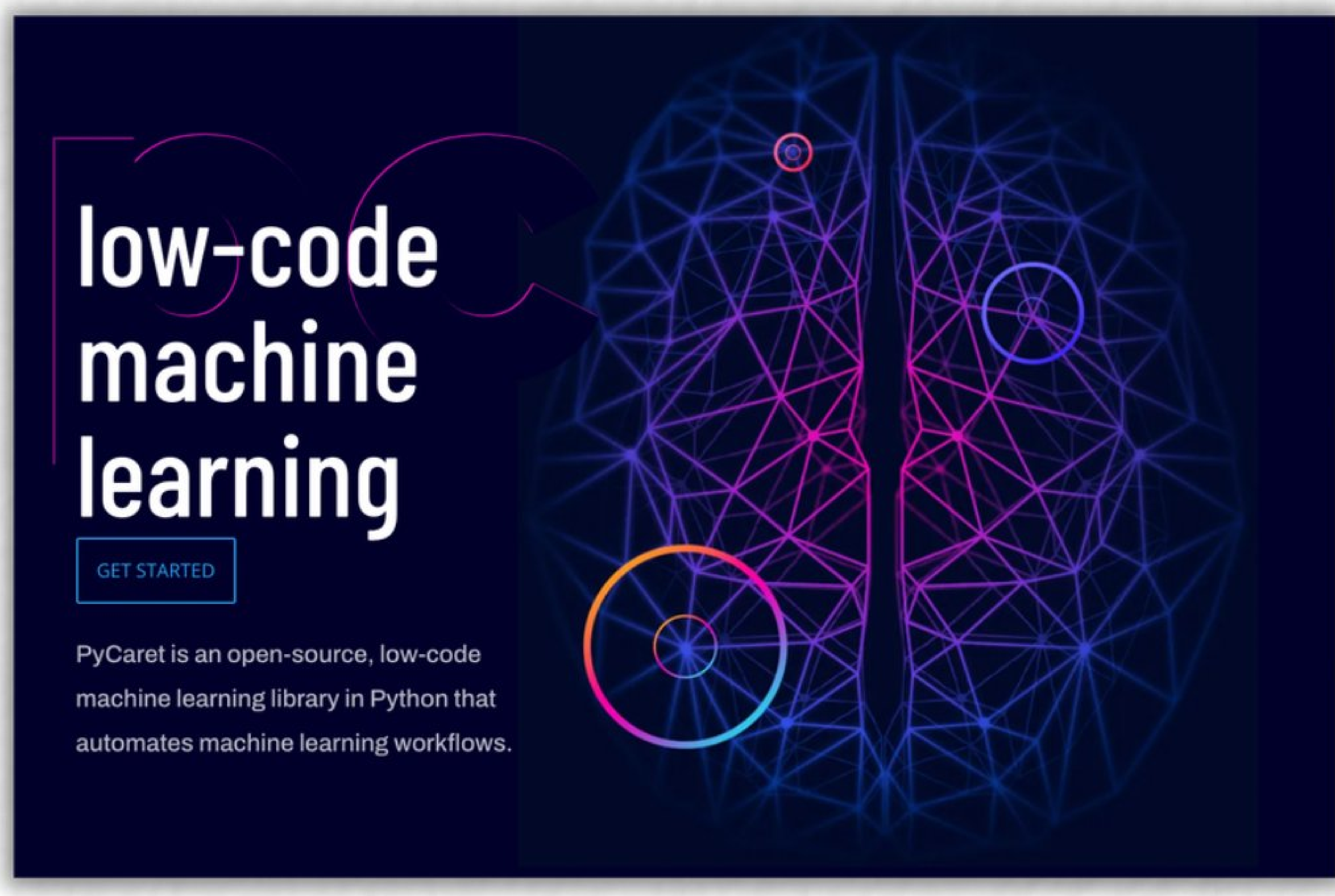
Here are **30 open-source libraries** to upgrade your data game.

# 1. YellowBrick




A suite of visualization and diagnostic tools for **faster model selection**.

# 2. PyCaret




**Automate ML workflows** with this **low-code library**.

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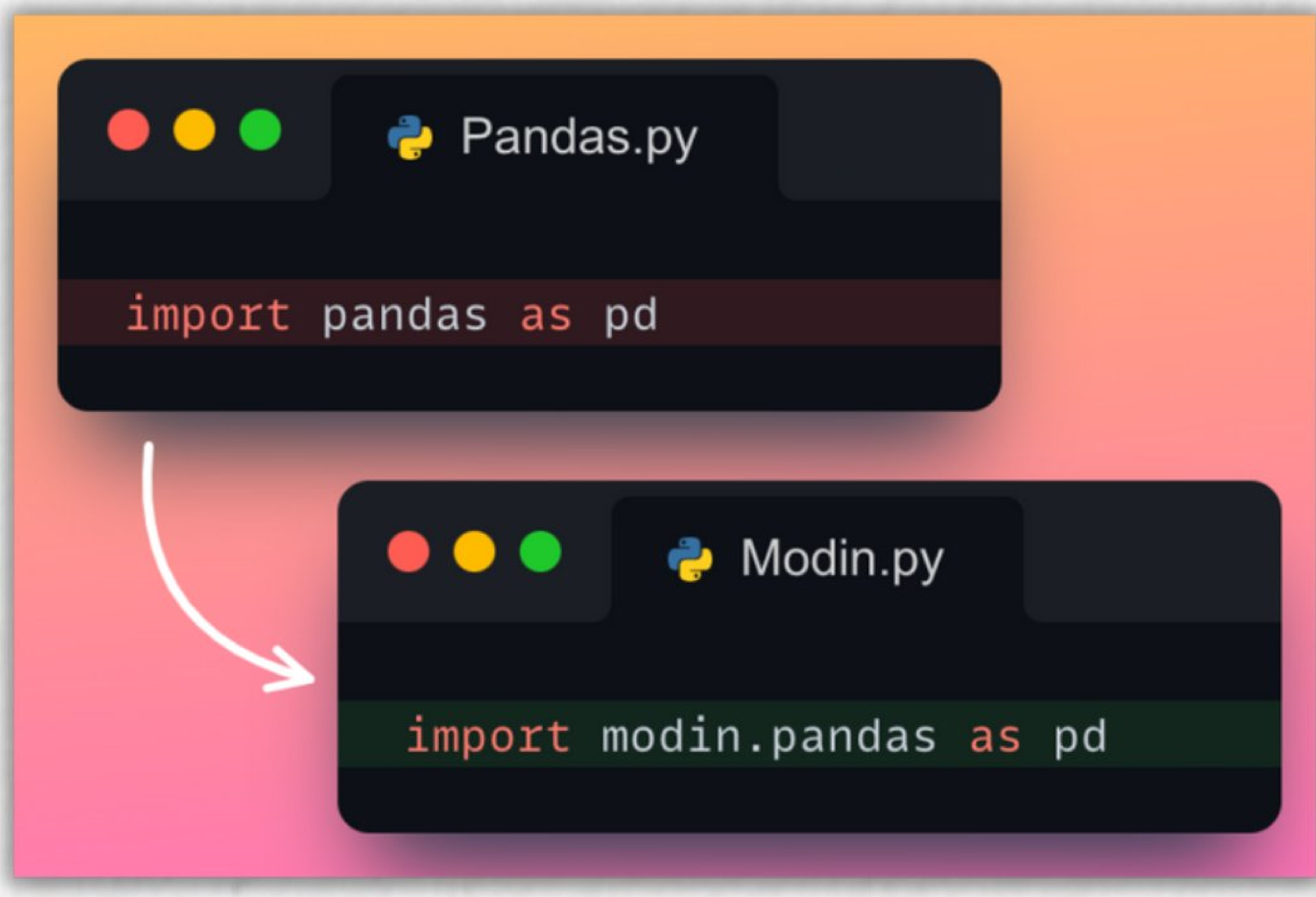
# 3. imbalanced-learn

```
target.value_counts()
1         548
0         352
```

```
from imblearn.over_sampling import SMOTE

os = SMOTE()

X_new, y_new = os.fit_resample(data, target)

y_new.value_counts()
1         548
0         548
```

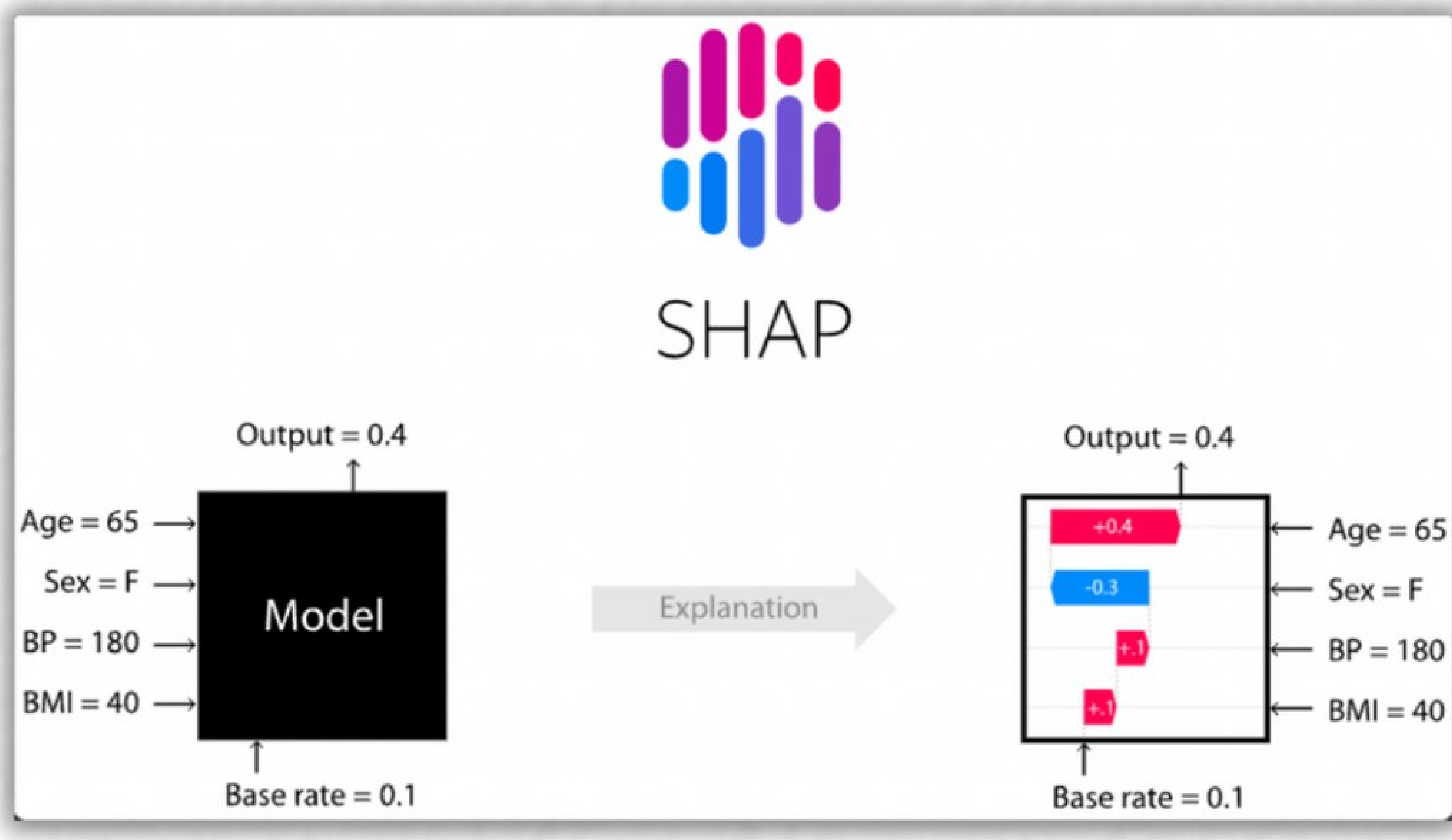
A variety of methods to handle class imbalance.

# 4. Modin



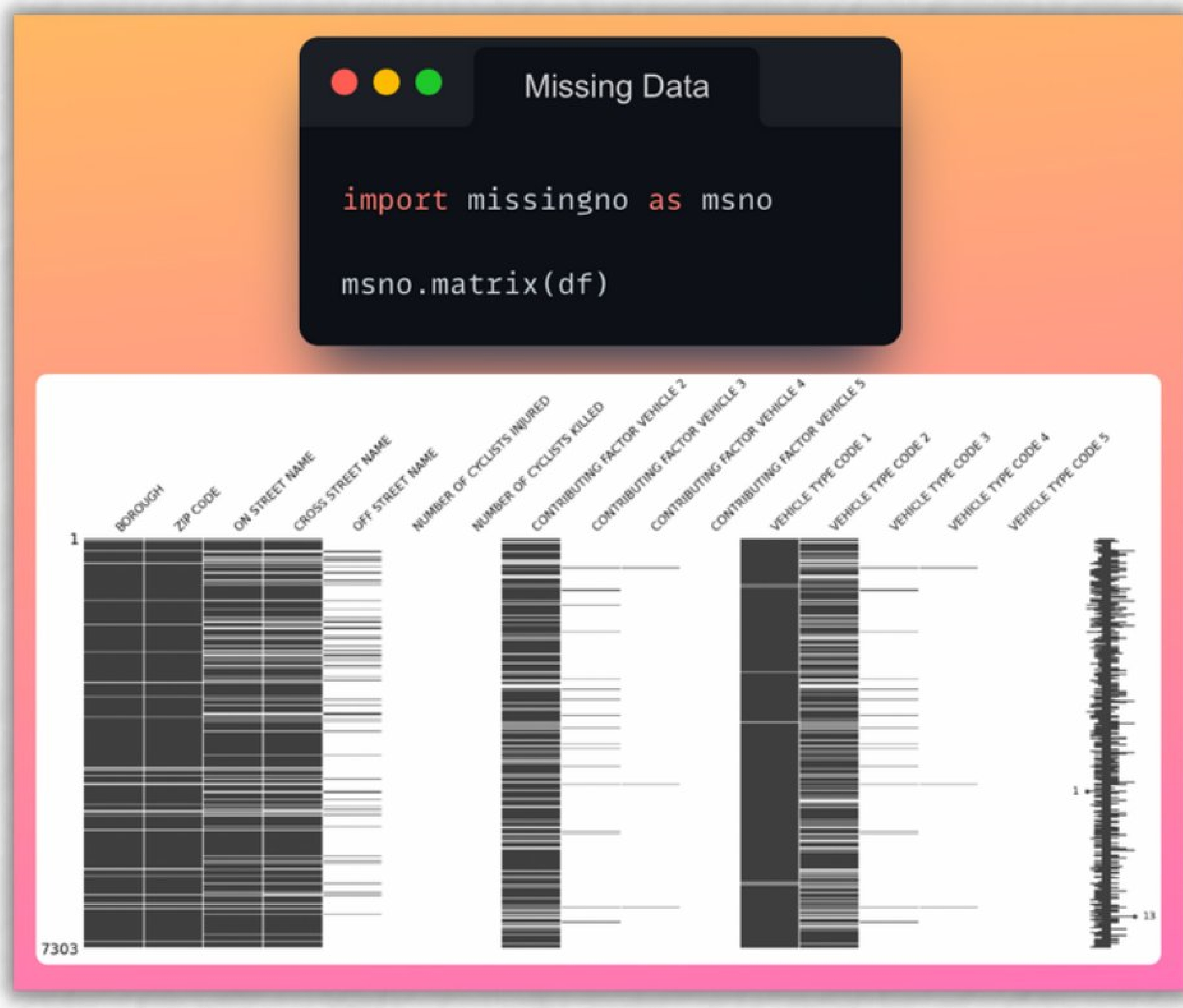

Boost Pandas' performance **up to 70x** by modifying the import.

# 5. SHAP



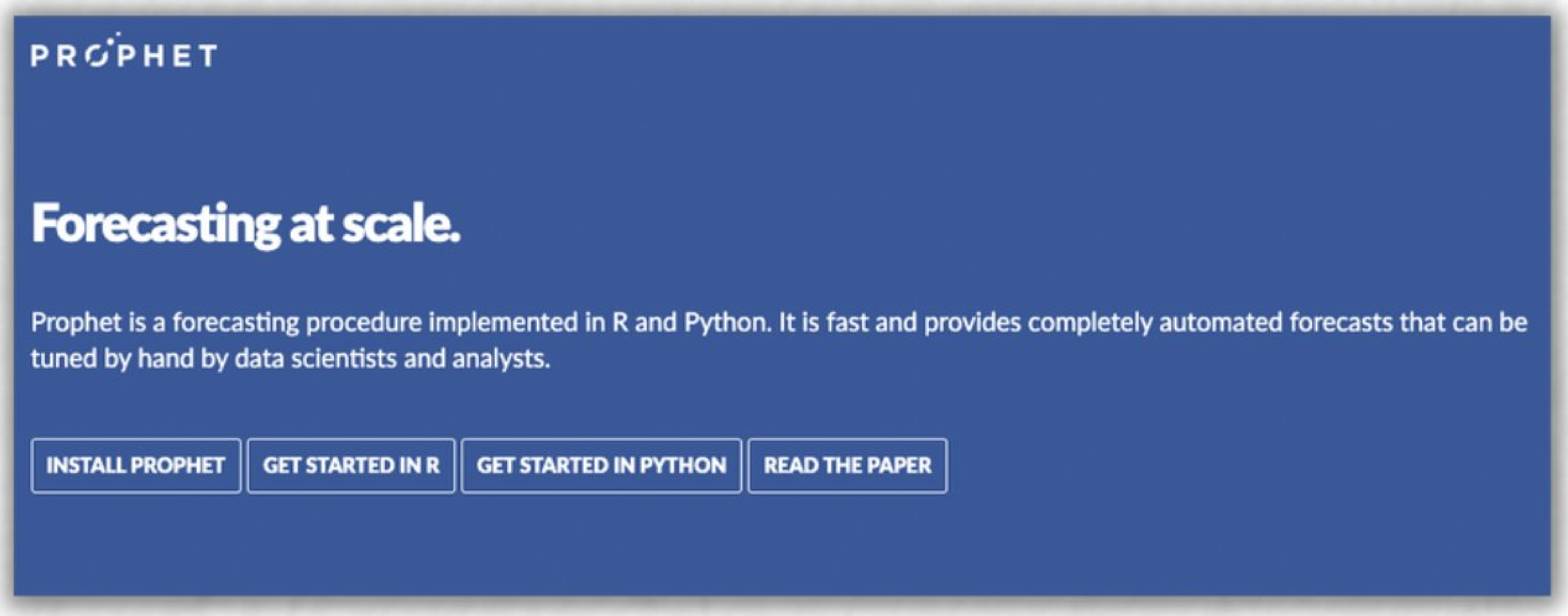

**Explain the output of any ML model in few lines of code.**

# 6. Missingno



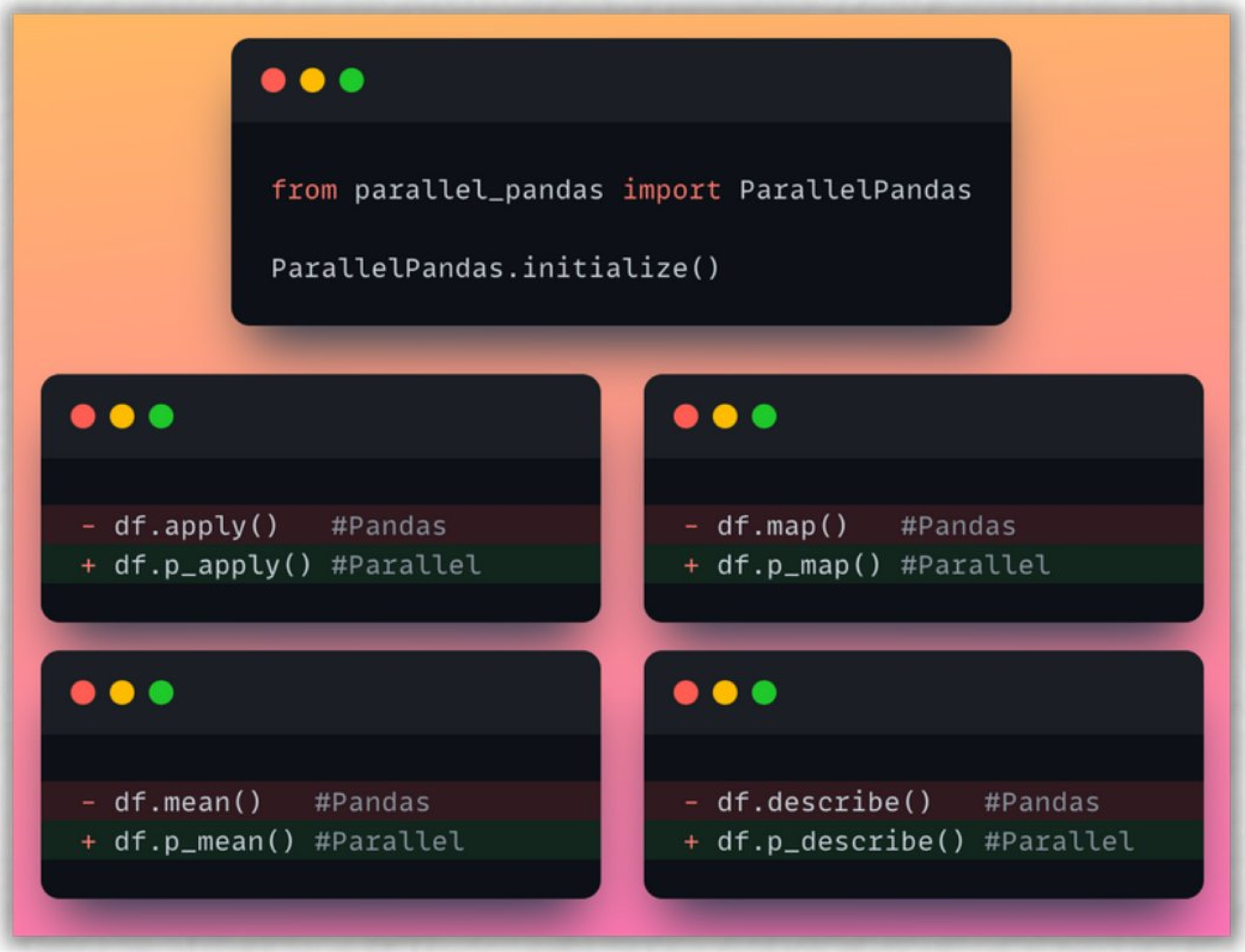

**Visualize missing values** in your dataset with ease.

# 7. Prophet




Produce **high-quality forecasts** on time-series data.

# 8. Parallel-Pandas




Parallelize Pandas across all CPU cores for **faster computation**.

# 9. Featuretools


Automated feature engineering for ML models.

# 10. Lazy Predict

```python
from lazypredict.Supervised import LazyRegressor

reg = LazyRegressor()
reg.fit(X_train, X_test, y_train, y_test)
```

```
| Model                 | R-Squared |   RMSE   | Time Taken |
|:----------------------|----------:|---------:|-----------:|
| SVR                   | 0.877199  | 2.62054 |   0.0330021 |
| KNeighborsRegressor   | 0.826307  | 3.1166  |   0.0179954 |
| MLPRegressor          | 0.750536  | 3.73503 |   0.725997  |
                            ...
                            ...
| LinearRegression      | 0.71753   | 3.97444 |   0.0190051 |
| DummyRegressor        |-0.02157   | 7.55832 |   0.0140116 |
```

Train **30 machine learning models** in one line of code.

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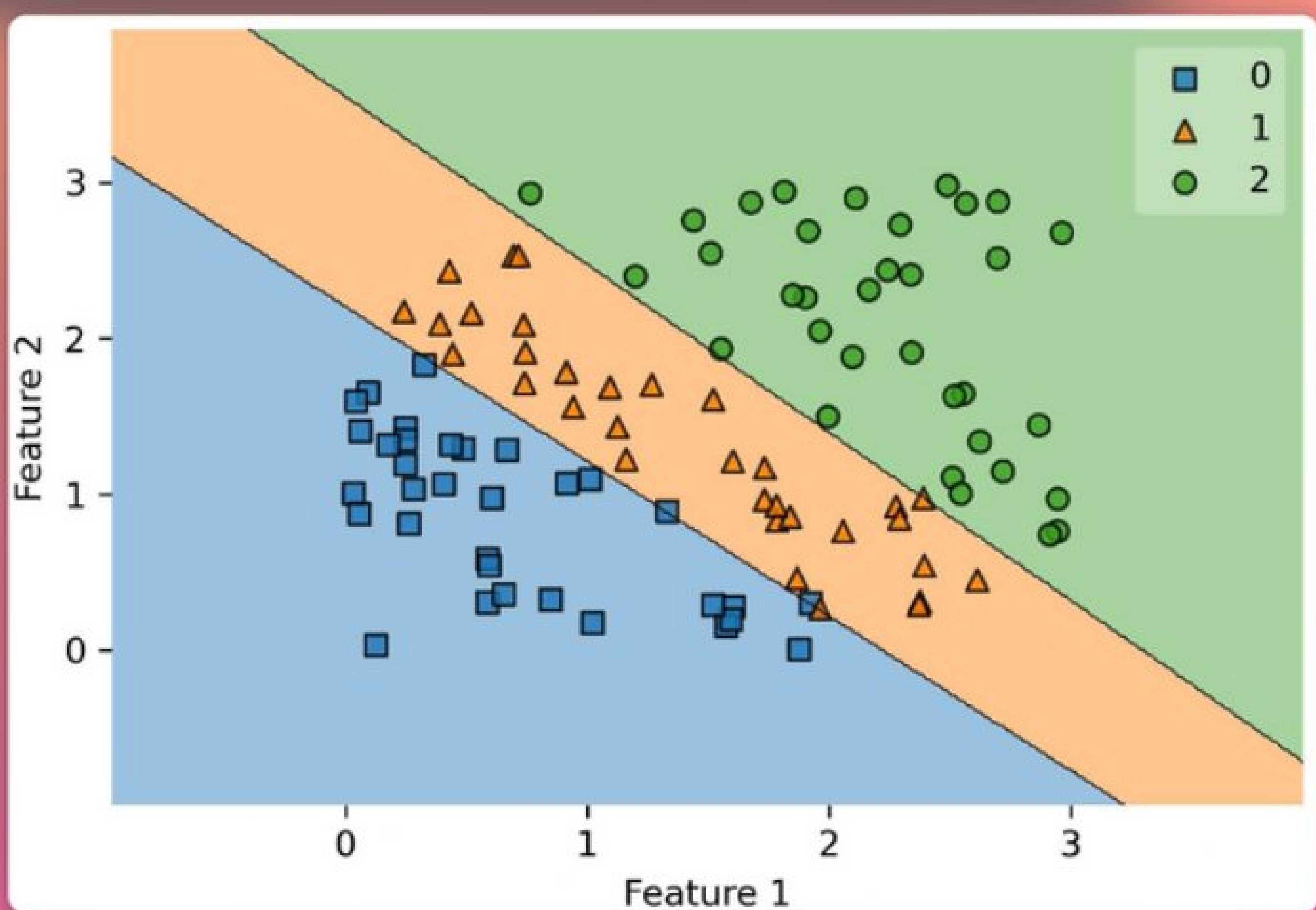
# 11. mlxtend

```python
from mlxtend.plotting import plot_decision_regions

model = LogisticRegression().fit(X, y)

plot_decision_regions(X, y, model)
```




A collection of utility functions for **processing, evaluating, visualizing** models.

# 12. Vaex

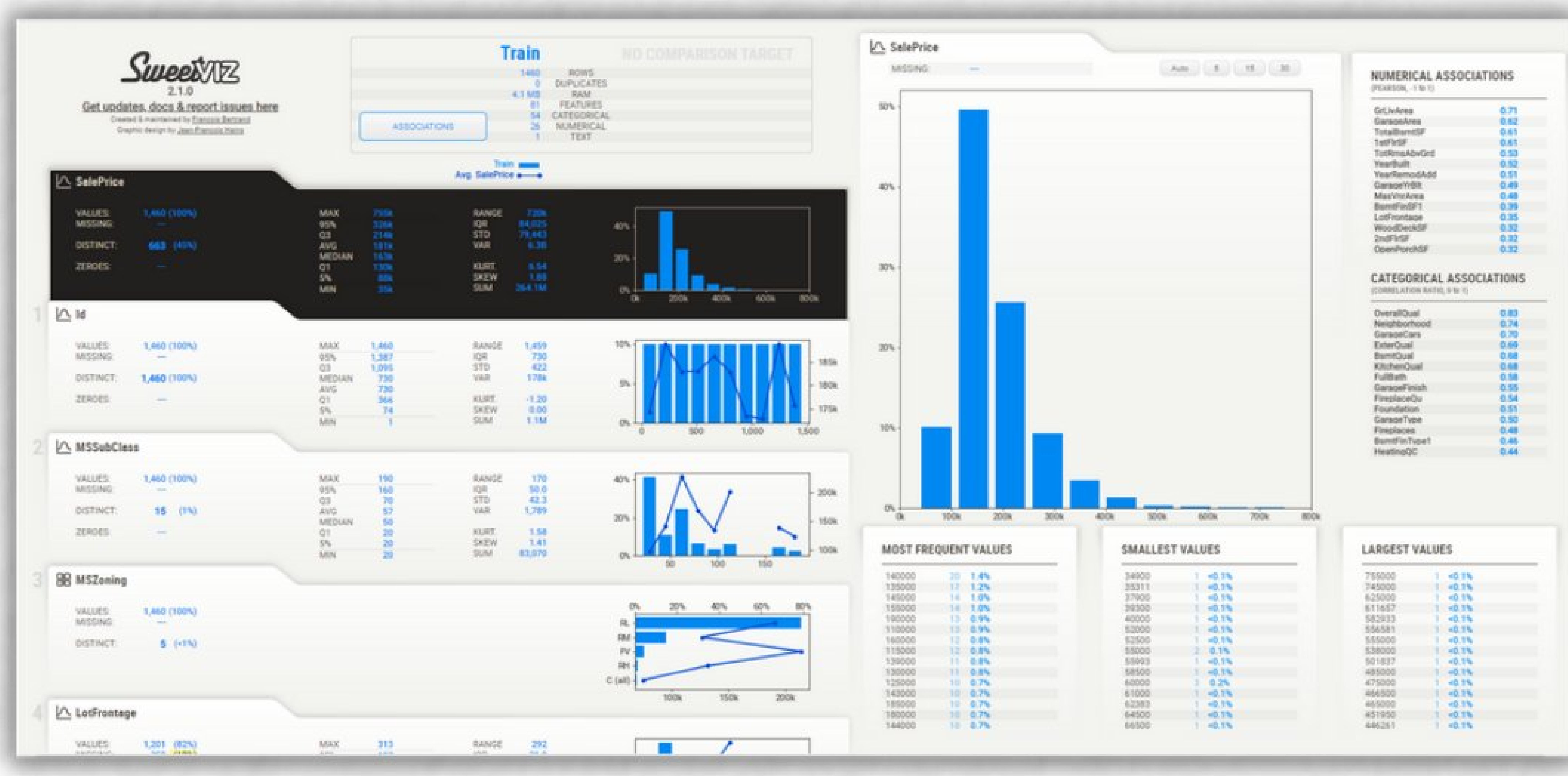
```
%%time
df
```

```
CPU times: user 8 µs, sys: 0 ns, total: 8 µs
Wall time: 16.7 µs
```

| # | vendor_id | pickup_datetime | dropoff_datetime | passenger_count | payment_type | trip_distance | pickup_longitude | pickup_latitude |
|---|---|---|---|---|---|---|---|---|
| 0 | VTS | 2009-01-04 02:52:00.000000000 | 2009-01-04 03:02:00.000000000 | 1 | CASH | 2.630000114440918 | -73.99195861816406 | 40.72156524658203 |
| 1 | VTS | 2009-01-04 03:31:00.000000000 | 2009-01-04 03:38:00.000000000 | 3 | Credit | 4.550000190734863 | -73.98210144042969 | 40.736289978027344 |
| 2 | VTS | 2009-01-03 15:43:00.000000000 | 2009-01-03 15:57:00.000000000 | 5 | Credit | 10.350000381469727 | -74.0025863647461 | 40.73974609375 |
| 3 | DDS | 2009-01-01 20:52:58.000000000 | 2009-01-01 21:14:00.000000000 | 1 | CREDIT | 5.0 | -73.9742660522461 | 40.79095458984375 |
| 4 | DDS | 2009-01-24 16:18:23.000000000 | 2009-01-24 16:24:56.000000000 | 1 | CASH | 0.4000000059604645 | -74.00157928466797 | 40.719383239746094 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 1,173,057,922 | VTS | 2015-12-31 23:59:56.000000000 | 2016-01-01 00:08:18.000000000 | 5 | 1 | 1.2000000476837158 | -73.99381256103516 | 40.72087097167969 |
| 1,173,057,923 | CMT | 2015-12-31 23:59:58.000000000 | 2016-01-01 00:05:19.000000000 | 2 | 2 | 2.0 | -73.96527099609375 | 40.76028060913086 |
| 1,173,057,924 | CMT | 2015-12-31 23:59:59.000000000 | 2016-01-01 00:12:55.000000000 | 2 | 2 | 3.799999952316284 | -73.98729705810547 | 40.739078521728516 |
| 1,173,057,925 | VTS | 2015-12-31 23:59:59.000000000 | 2016-01-01 00:10:26.000000000 | 1 | 2 | 1.9600000381469727 | -73.99755859375 | 40.72569274902344 |
| 1,173,057,926 | VTS | 2015-12-31 23:59:59.000000000 | 2016-01-01 00:21:30.000000000 | 1 | 1 | 1.059999942779541 | -73.9843978881836 | 40.76725769042969 |

High performance package for **lazy Out-of-Core DataFrames.**

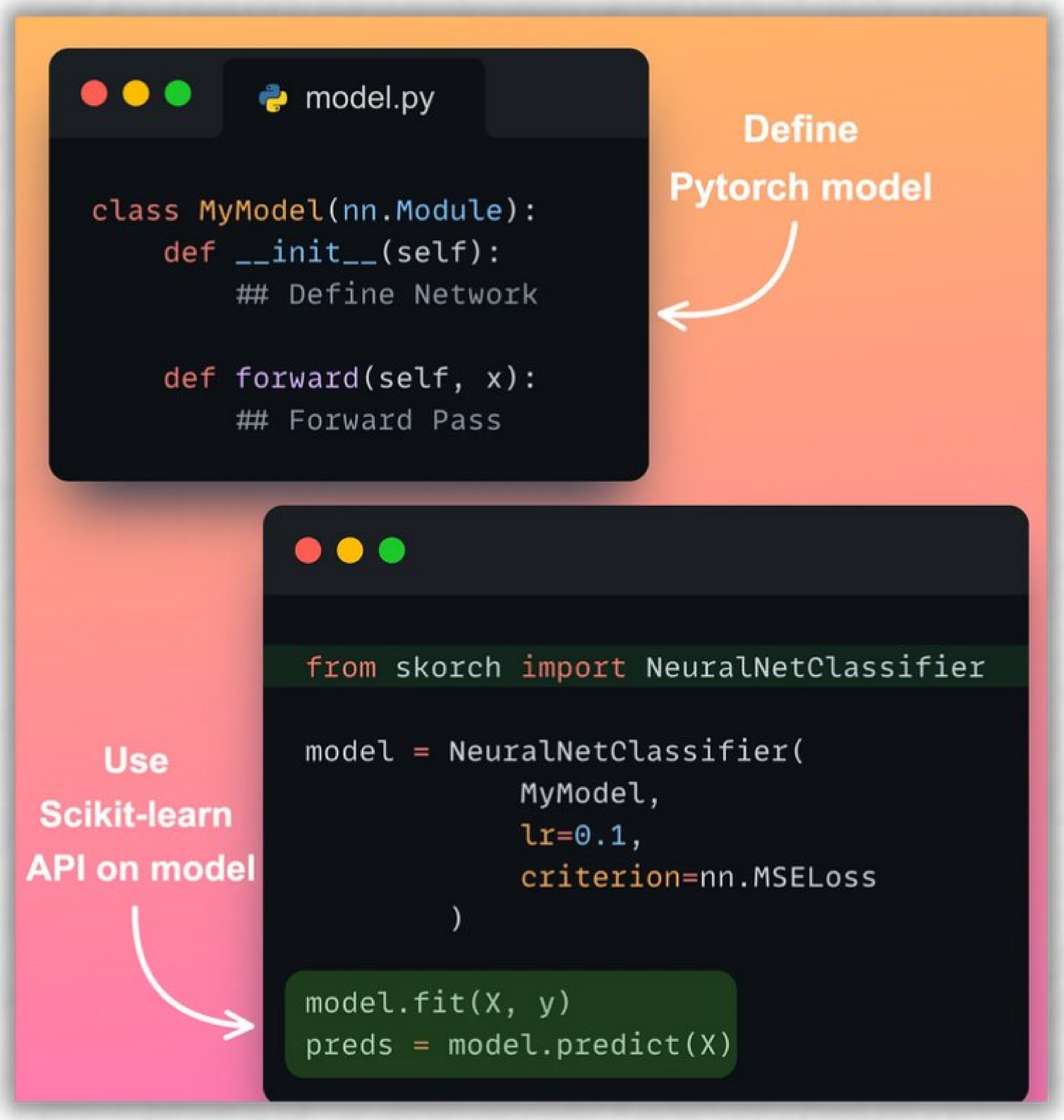
# 13. SweetViz




**In-depth EDA report**

in two lines
of code.

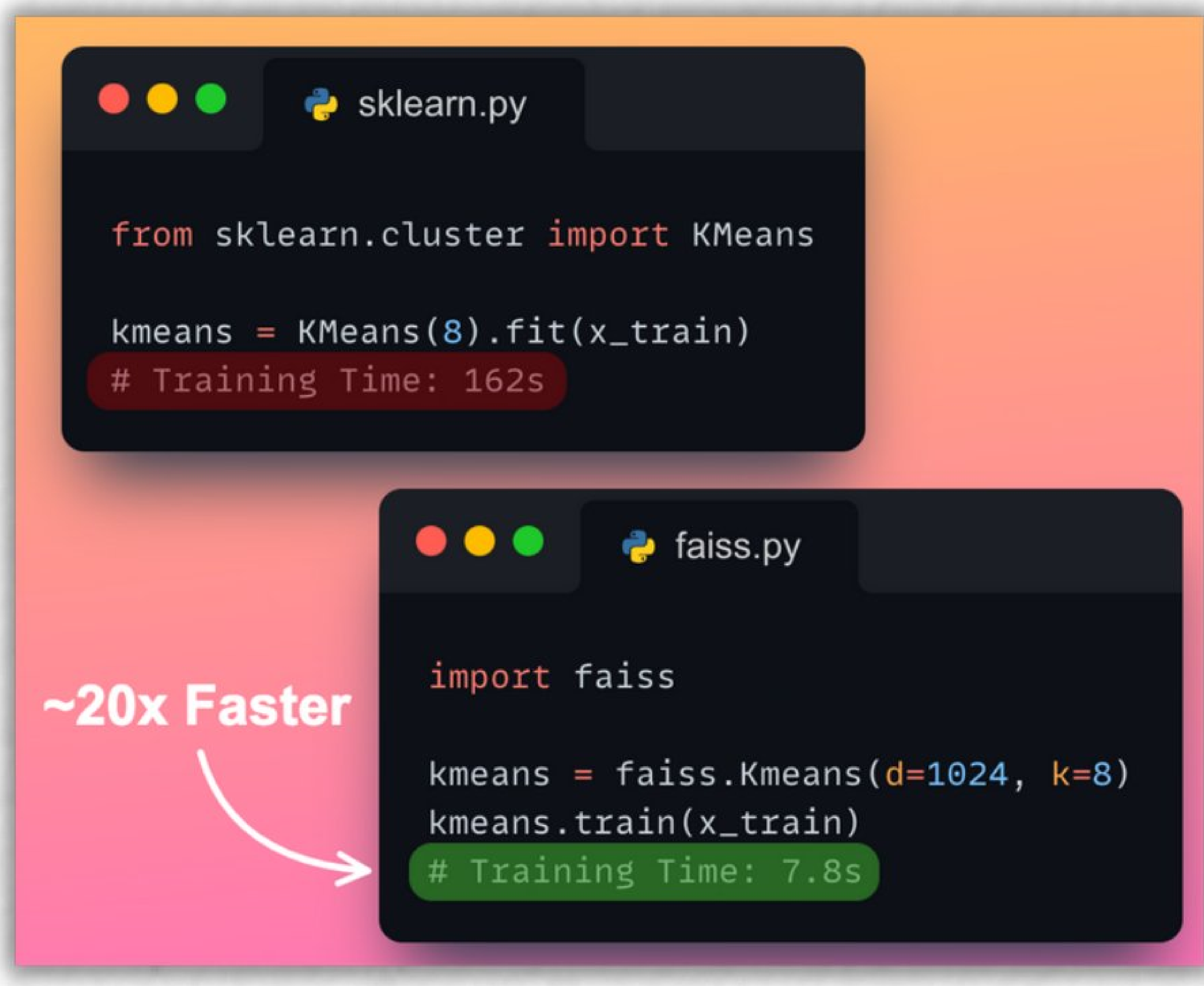
# 14. Skorch




PyTorch Sklearn

Leverage the **power of PyTorch** with the **elegance of sklearn**.

# 15. Faiss




Efficient algorithms for **similarity search** and **clustering** dense vectors.

# 16. statsmodel

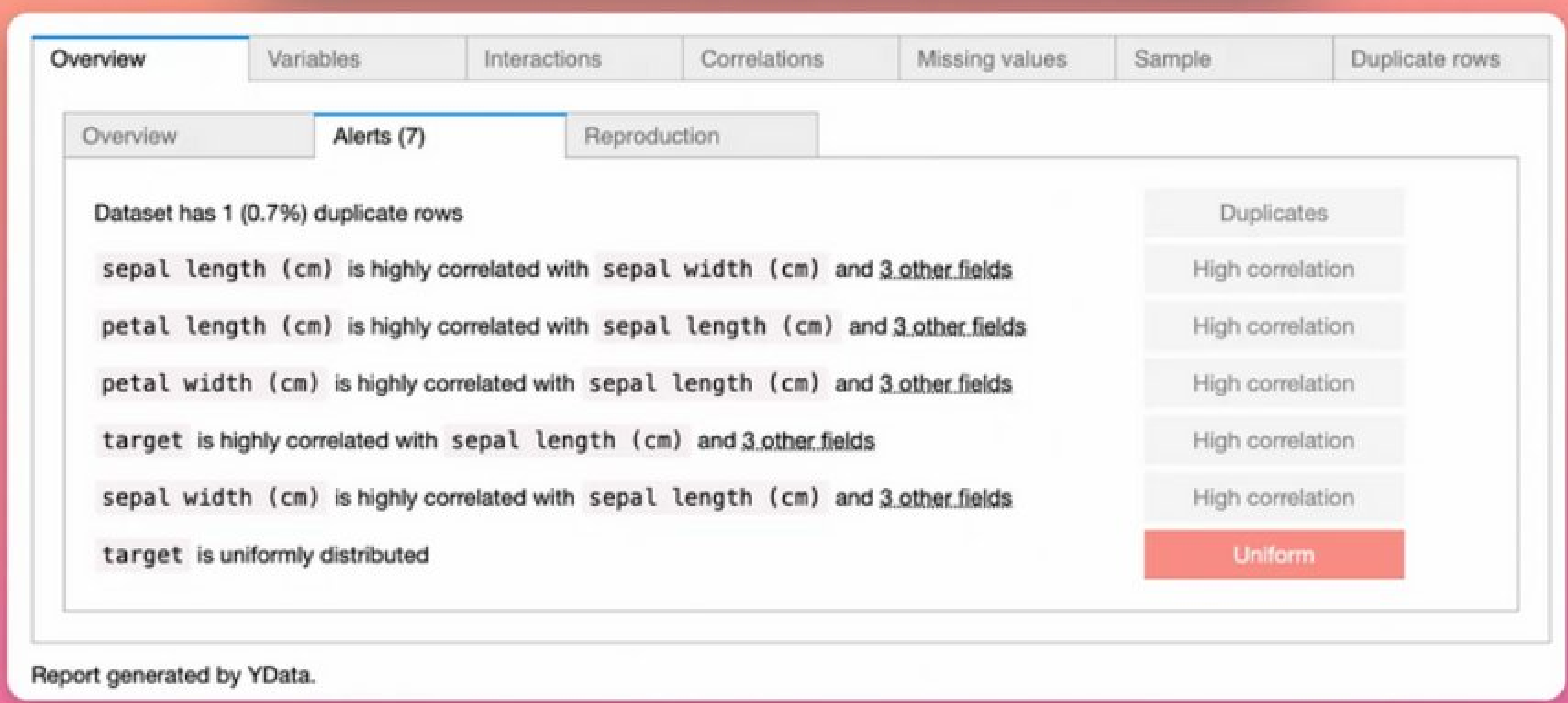
```python
import statsmodels.api as sm

results = sm.OLS(y, X).fit()
print(results.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.178
Model:                            OLS   Adj. R-squared:                  0.161
Method:                 Least Squares   F-statistic:                     10.51
Date:                Wed, 02 Nov 2022   Prob (F-statistic):           7.41e-05
Time:                        17:12:45   Log-Likelihood:                -20.926
No. Observations:                 100   AIC:                             47.85
Df Residuals:                      97   BIC:                             55.67
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          1.4713      0.075     19.579      0.000       1.322       1.620
x1             0.1045      0.105      1.000      0.320      -0.103       0.312
x2             0.4831      0.107      4.503      0.000       0.270       0.696
==============================================================================
Omnibus:                       39.684   Durbin-Watson:                   1.848
Prob(Omnibus):                  0.000   Jarque-Bera (JB):                6.503
Skew:                           0.096   Prob(JB):                       0.0387
Kurtosis:                       1.766   Cond. No.                         5.09
==============================================================================
```

Statistical testing and data exploration at fingertips.

# 17. Pandas-Profiling

```python
from pandas_profiling import ProfileReport

profile = ProfileReport(iris_data,
                        title="Pandas Profiling Report")

profile.to_widgets()
```




Generate a high-level **EDA report** of your data in no time.

# 18. Streamlit


**Create and host data-based Python web apps**
in few lines
of code.

# 19. Category-encoders

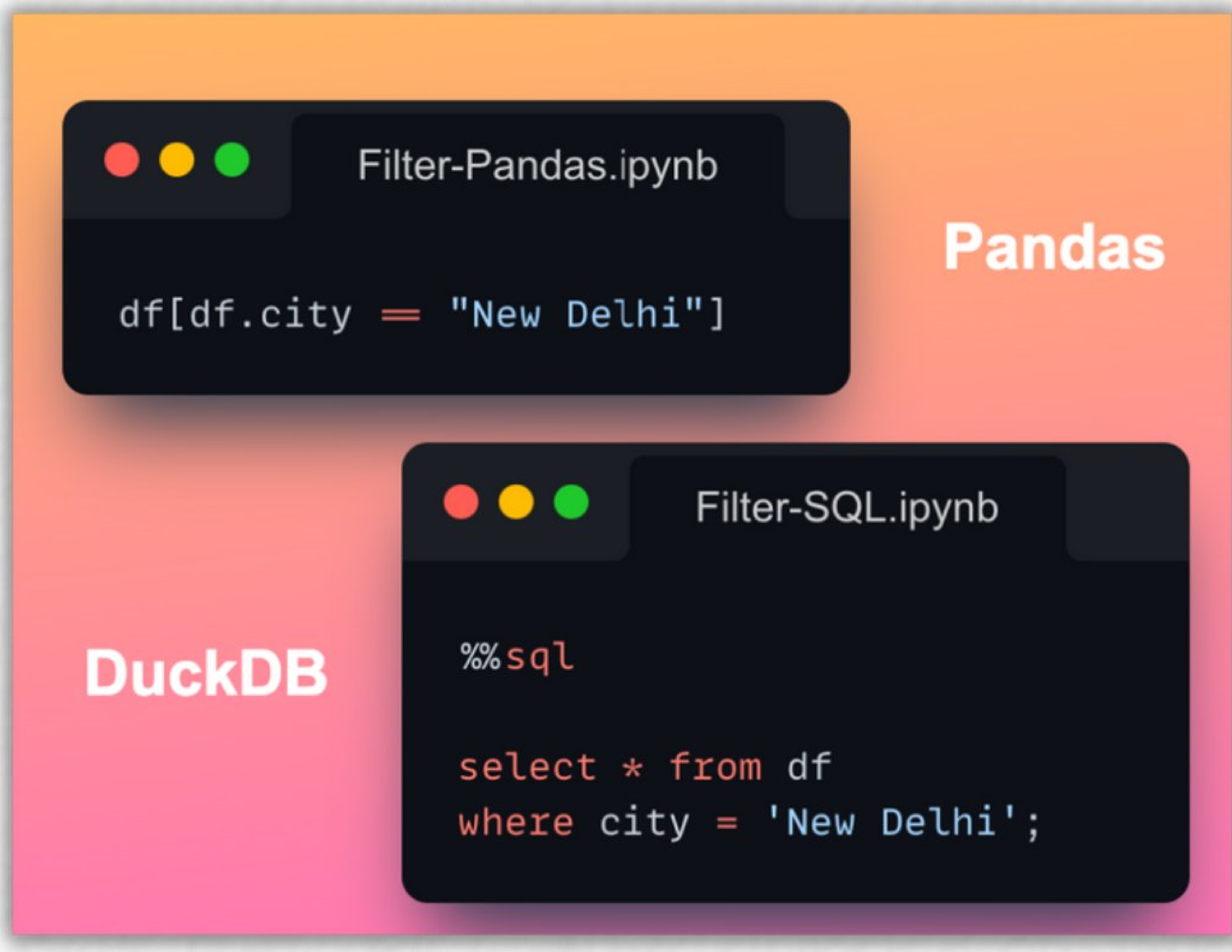
```python
import category_encoders as ce

ce_be = ce.BinaryEncoder(cols=['class'])

ce_be.fit_transform(data["class"])

   class_0  class_1  class_2
0        0        0        1
1        0        1        0
2        0        1        1
3        1        0        0
4        0        0        1
```

Binary

| | gender | class |
|---|---|---|
| 0 | Male | A |
| 1 | Female | B |
| 2 | Male | C |
| 3 | Female | D |
| 4 | Female | A |

Over 15 categorical data encoders.

# 20. DuckDB




Run **SQL queries on DataFrame**.

# 21. PandasML

```python
import pandas_ml as pdml

df = pdml.ModelFrame(X, y)

train, test = df.model_selection.
               train_test_split()

reg = df.linear_model.LinearRegression()

train.fit(reg)

test.predict(reg)
```

**Pandas data wrangling + Sklearn algorithms + Matplotlib visualization.**

# 22. Pytest


An elegant testing framework to test your code.

# 23. Numexpr

```
import numpy as np
import numexpr as ne
```

```
a = np.random.random(10**7)
b = np.random.random(10**7)
```

```
%timeit np.cos(a) + np.sin(b)
```

```
142 ms ± 257 µs per loop
```

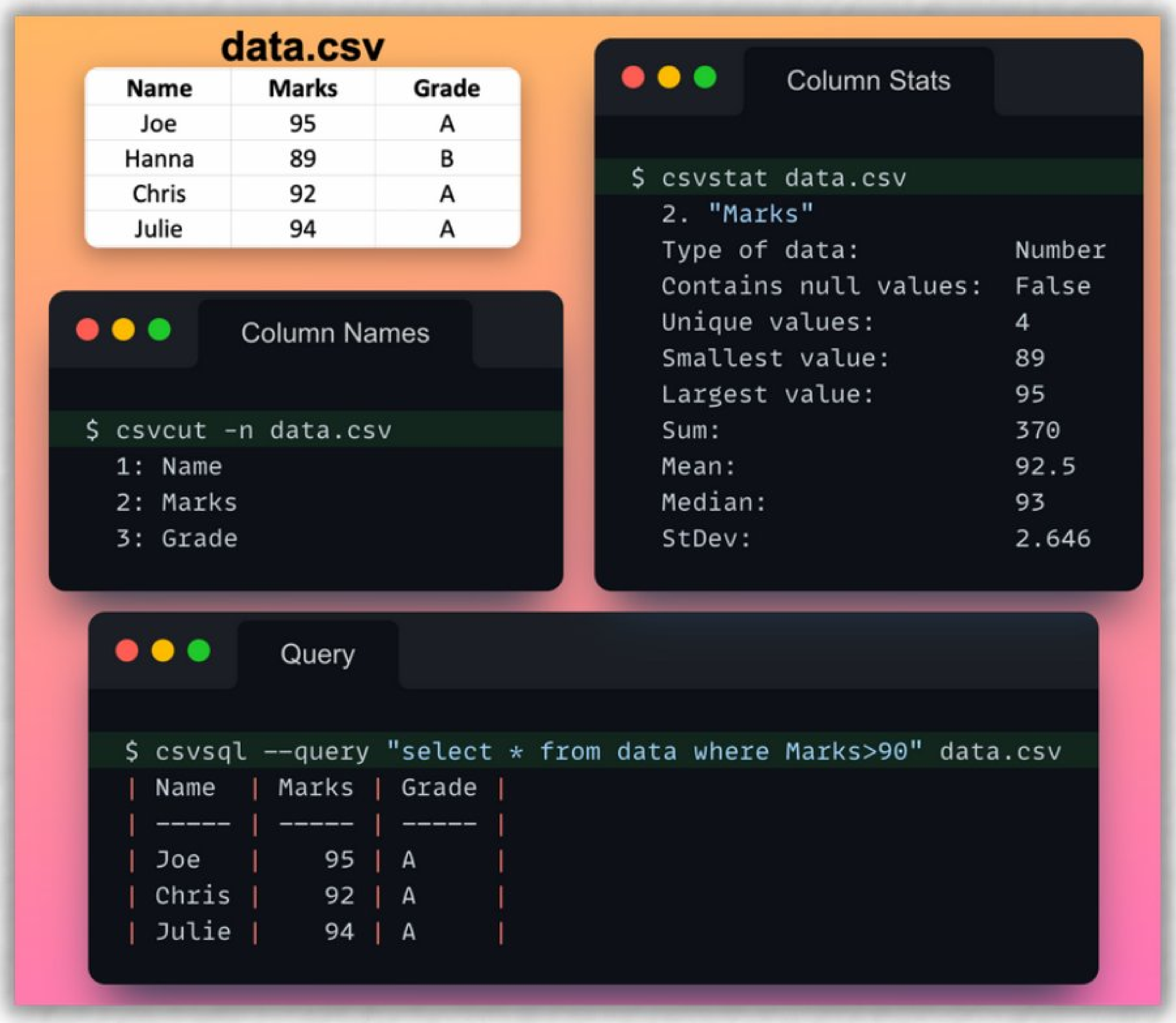
```
%timeit ne.evaluate("cos(a) + sin(b)")
```

```
32.5 ms ± 229 µs per loop
```

**~5x Faster**

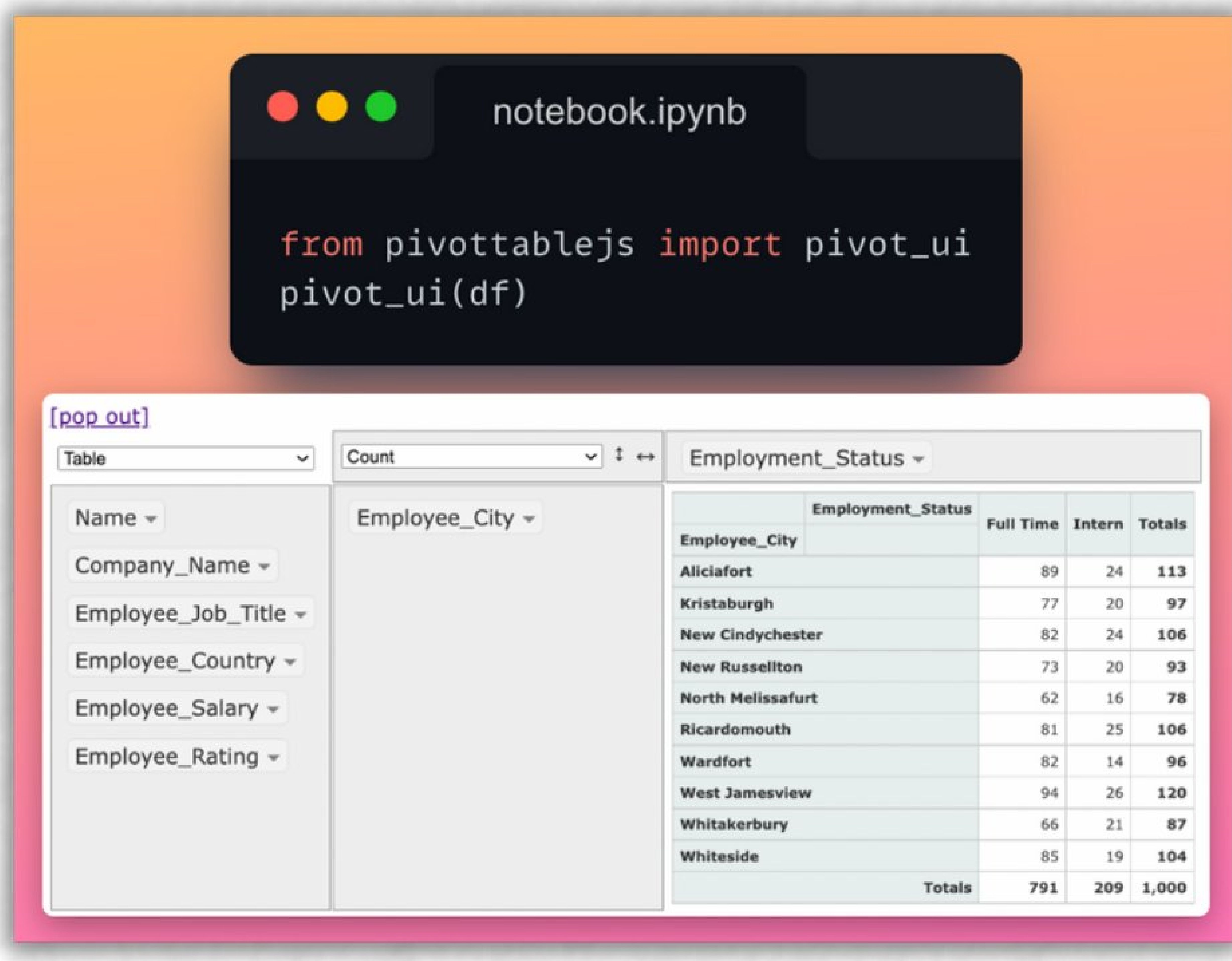
Parallelize NumPy to all CPU cores for **20x speedup**.

# 24. CSV-Kit




Explore, query and describe CSV files from terminal.

# 25. PivotTableJS




```python
from pivottablejs import pivot_ui
pivot_ui(df)
```

| Employee_City \ Employment_Status | Full Time | Intern | Totals |
|---|---|---|---|
| Aliciafort | 89 | 24 | 113 |
| Kristaburgh | 77 | 20 | 97 |
| New Cindychester | 82 | 24 | 106 |
| New Russellton | 73 | 20 | 93 |
| North Melissafurt | 62 | 16 | 78 |
| Ricardomouth | 81 | 25 | 106 |
| Wardfort | 82 | 14 | 96 |
| West Jamesview | 94 | 26 | 120 |
| Whitakerbury | 66 | 21 | 87 |
| Whiteside | 85 | 19 | 104 |
| Totals | 791 | 209 | 1,000 |

**Drap-n-drop** tools to group, pivot, plot dataframe.

# 26. Faker

fake_data.py

```python
from faker import Faker
fake = Faker()

>>> fake.name()
'Darrell Alexander'

>>> fake.email()
'ryanrichard@example.com'

>>> fake.address()
'205 Brown Point, West Melissaport, MN 93828'
```

Generate **fake yet meaningful data** in seconds.

# 27. Icecream

Print

```python
def func(arr, n):
    print("arr =", arr, "n =", n)

func([1,2,3], 2)
## arr = [1, 2, 3] n = 2
```

Icecream

```python
from icecream import ic

def func(arr, n):
    ic(arr, n)

func([1,2,3], 2)
## ic| arr: [1, 2, 3], n: 2
```

**Don't debug with print()**. Use icecream.

# 28. Pyforest

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import sys

from sklearn.linear_model import LinearRegression
```

```python
from pyforest import *

pd.read_csv("file.csv") ✅
np.array([1,2,3])       ✅
sys.path                ✅
LinearRegression()      ✅
```

No need to write imports.
**Automatic package import.**

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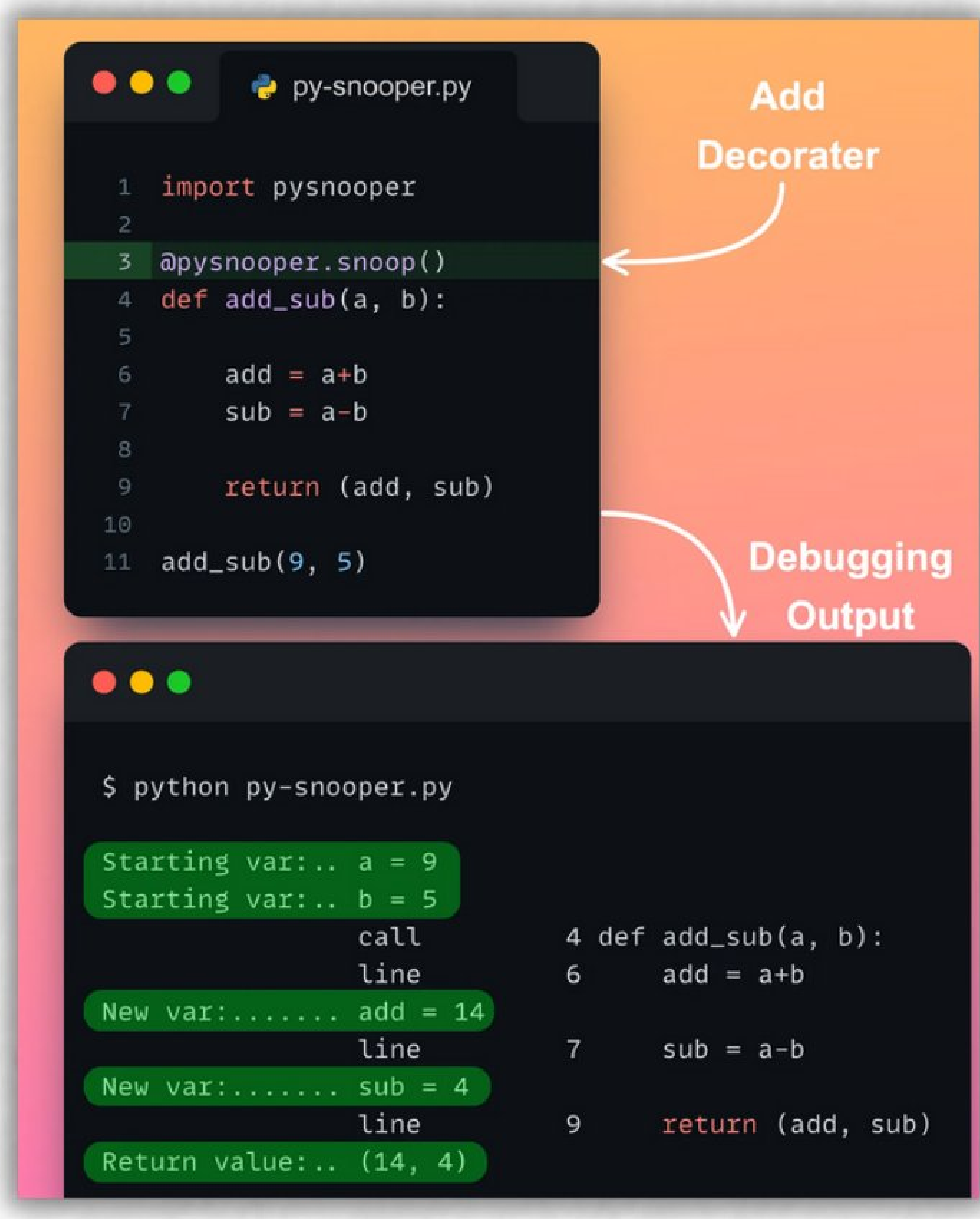
# 29. PySnooper




Profile your code. **Track new variables, and their updates**.

# 30. Sidetable

sidetable.py

```python
import sidetable

df.stb.freq(["City"],
            style=True)
```

**value_counts()**

```
City
West Jamesview       120
Aliciafort           113
New Cindychester     106
Ricardomouth         106
Whiteside            104
Kristaburgh           97
Wardfort              96
New Russellton        93
Whitakerbury          87
North Melissafurt     78
```

| | City | count | percent | cumulative_count | cumulative_percent |
|---|---|---|---|---|---|
| 0 | West Jamesview | 120 | 12.00% | 120 | 12.00% |
| 1 | Aliciafort | 113 | 11.30% | 233 | 23.30% |
| 2 | Ricardomouth | 106 | 10.60% | 339 | 33.90% |
| 3 | New Cindychester | 106 | 10.60% | 445 | 44.50% |
| 4 | Whiteside | 104 | 10.40% | 549 | 54.90% |
| 5 | Kristaburgh | 97 | 9.70% | 646 | 64.60% |
| 6 | Wardfort | 96 | 9.60% | 742 | 74.20% |
| 7 | New Russellton | 93 | 9.30% | 835 | 83.50% |
| 8 | Whitakerbury | 87 | 8.70% | 922 | 92.20% |
| 9 | North Melissafurt | 78 | 7.80% | 1,000 | 100.00% |

Supercharge Pandas' **value_counts()** method.

# Hope that helped.

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